Raw Farmhouse Harvest Ale Recipe (~280L)

Brewery: Keldling Brew

Brew: #187

Beer Title: 'Pick Me!'

Batch Specifications

- **Final Volume:** ~280 Liters
- **Kettle Volume:** 190 Liters
- **Target Original Gravity (OG):** 1.055
- **Target Final Gravity (FG):** 1.009 - 1.011
- **Estimated ABV:** 5.8% - 6.1%
- **Process Type:** Raw Ale (No-Boil, 80°C Hop Stand)

Grain Bill & Sugars

- 25.0 kg Golden Promise Malt
- 25.0 kg Maris Otter Malt
- 5.0 kg Caramel Malt (50 EBC)
- 5.0 kg Chit Malt
- 5.0 kg Cane Sugar (Added into the boil kettle)
- Additional cane sugar used within the yeast starters

Hop Bill

- 500g Chinook Pellets (half as mash hops, half boiled separately in 2L water)
- 500g Fuggle (Home-Dried Whole Leaf)
- 500g Willamette (Home-Dried Whole Leaf)
- 500g Styrian Golding (Home-Dried Whole Leaf)
- 500g Bramling Cross (Home-Dried Whole Leaf)
- 500g Phoenix (Home-Dried Whole Leaf)

Water Adjustments & pH Control

- Lime juice, Lemon juice, and/or Citric Acid (To hit target transfer pH)

Yeast & Nutrients

- Co-pitch: Hornindal Kveik + Ebbegarden Kveik
- Servomyces Yeast Nutrient (Added directly to the unitank)

Brew Process

1. Hop Preparation (Day Before)

- Harvest wet hop cones in the morning after dew evaporates.
- Dry the wet hops in the food dehydrator at 35°C. Note the hop weight per plate.
- Dry for 10 hours or more until the dry hop weight is 20%-30%.

2. Yeast Starter (Day Before)

- Do a 5L starter per yeast with some yeast nutrients and 125gr cane sugar pr. Liter.

3. Tank Work

- Fill hot liquor kettle and boil kettle with water, adjust the water pH to below 5.5 and heat it to 80°C.
- Let CO_2 of the previous brew out, empty and clean the tank.
- Dismantle parts, clean them and put them back.
- Transfer the hot water to the tank to disinfect it. Let it sit.

4. Mashing

- Fill/Refill your brew system with water from the tank when you need it.
- Perform your standard mashing using the malts.
- Add half of Chinook pellets as mash hops in a spider/bag.
- Transfer the sweet wort into the boil kettle.

5. The Bittering Tea

- On the side, boil 2 L water.
- Add 250g of Chinook pellets.
- Let it just cool naturally for 15 min.

6. The Dilution/Disinfection Reservoir

- After mashout refill the hot liquor kettle with water from the tank.
- Heat it to 80°C.
- pH-adjust it to 5.0.

7. More Tank work

- Let any remaining water out of the Tank.
- Set the max temperature to 35°C (check that the tank still has a port open!)

- Let the chiller bring the tank temperature down.

8. The 80°C Hop Stand & pH Adjustment

- Bring the boil kettle to 80°C. Do not boil (I know you want to).
- Add the bittering tea and cane sugar.
- Bring over the mash hops in its hop spider/bag for more hop extraction.
- Maintain the temperature at 80°C for 1 hour to pasteurize the wort and extract volatile oils. Re-circulate to keep temperature and gently motion the wort.
- Add all of your home-dried free-floating whole-leaf hops into the kettle and let them sit for at least 10-30 min. Keep re-circulating for motion to extract Lupulin.
- **pH Control Step:** After hop stand measure the wort pH. Add Lime juice, Lemon juice, and/or Citric Acid to drive the kettle pH down to exactly **4.9** before starting the transfer. The target pH in the beer is 4.5 or below.

9. Transfer, Inline Chilling, and Dilution

- Route the plumbing: Hot Liquor Kettle Outlet → HopRocket (Empty acting as a strainer) → Pump → Plate Chiller → Drain.
- Disinfect the plumbing by running hot water through it.
- ReRoute the plumbing: Boil Kettle Outlet → HopRocket (Empty acting as a strainer) → Pump → Plate Chiller → Unitank.
- Start plate chiller chilling loop (I recirculate water from a ground cooled reservoir).
- Begin pumping the clear wort through the plate chiller to drop the temperature immediately to your pitching target (30°C–35°C) on its way to the unitank.
- As the liquid level drops in the boil kettle, slowly rinse the hop bed using your extra boiled dilution water to sweep remaining sugars through the plate chiller.
- Dilute the final wort volume in the unitank to roughly 280 Liters (adjusting dynamically to land on your target 1.055 OG).

10. Fermentation

- Confirm the unitank wort temperature is stable at 30°C–35°C.
- Add the Servomyces nutrient in a little boiled water directly into the unitank.
- Pitch the Yeast Starters.
- Maintain a fermentation temperature max of 35°C, if you like to, or disable the temperature controller. When the yeast takes off the temperature will rise further, and then the temperature will slowly fall to room temperature with time.
- If you want to capture CO2, be on your toes; it might just need a day or 2 to ferment.
- After 3 days with a stable final gravity, cold-crash and force carbonate until you are pleased with its carbonation level.

Cheers,
Keld